

DESIGN, CONSTRUCTION, AND PERFORMANCE IN OHIO

Designing and building a house enabled me to put into practice many of the building science and engineering concepts and principles that I cover in my courses.

BY ALLEN ZIMMERMAN

My wife, Susan Choma, and I built our home in Ohio in 2000. I am a college professor, and my teaching responsibilities include engineering technology coursework in the areas of residential mechanical systems and energy-efficient construction. Designing and building the house enabled me to put into practice and observe the results of many of the building science and engineering concepts and principles that I cover in my courses.

Our overall goal was to build a home that was comfortable, safe, healthy, durable, quiet, low maintenance, practical, energy efficient, compatible with the site, affordable, and aesthetically appropriate. We considered all of these criteria when making trade-offs and decisions concerning size, design, materials, and equipment.

Location, Location, Location

The building site is in north central Ohio. It consists of a 6-acre meadow on the top of a ridge that overlooks several miles of wooded or farmed hills and valleys.

Natural gas is not available in the area. The electric supplier is a local co-op that provides electricity at very low rates, for three reasons. First, the co-op is member owned and managed as a nonprofit as opposed to being an investor-owned, for-profit utility. Second,



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Workers place concrete for a frost-protected shallow foundation monolithic footer/foundation/floor slab combination.

overhead costs are kept low by careful management, and by practices such as having members read their own meters and submit monthly meter readings. Third, the co-op is part of a consortium (of similar co-ops across Ohio) that is the power supplier to the member co-ops, and the consortium gets most of its power from two generating plants that it actually owns (one fueled by coal and the other a recently built natural gas-fired unit).

Design Details

Given the orientation of the house, the views, and energy considerations,

placing the garage on the north end of the structure was an obvious choice. We changed the typical ranch floor plan by placing the living and family rooms at the south end, where the maximum solar gain will occur, and moving the bedrooms to the north end adjacent to the garage. Efficient use of space is critical in a modest sized house such as this (2,165 ft² of conditioned space, consisting of 1,950 ft² in the main part of the house and a 215 ft² studio/laundry room in the garage section).

Window locations and sizes represent a compromise among various factors. These factors include code requirements, views, natural lighting, ventila-

tion, maximum solar gain in the winter (passive solar), minimum solar gain in the summer, performance, and aesthetics. Based on these factors, I selected a total window area in the main part of the house of approximately 15% of the floor area.

Our family does not accumulate and store very much “stuff,” nor do we find the use of basements as living areas appealing. Therefore, it made sense in terms of resource efficiency, energy efficiency, storage of passive-solar heat, cost, and minimum lot disturbance to choose a concrete floor instead of a basement or crawlspace. Raised perimeter shelving along all walls in the garage and a small, dedicated part of the garage floor area provide adequate space for stored items.

I specified raised-heel trusses with a wide overhang for a couple of reasons.

First, I wanted to overcome the problem of inadequate ceiling insulation in the eave areas adjacent to the top plates. The design of the trusses provides for a minimum of 12 inches of insulation at the plates. Second, I wanted an overhang that provides maximum shading of the south-facing windows in summer and winter. For this reason, I selected an overhang of 28 inches. The wide overhang also adds aesthetic appeal and provides additional protection for the windows, doors, and exterior walls.

(For more information on design, see “Energy Efficiency Features.”)

A Well-Insulated Building Envelope

Obviously, I intended to build a well-insulated and tight structure based

on the fundamentals of building science. However, when I took a holistic look at the various parameters involved, it became apparent that by upgrading the foundation, wall, and ceiling insulation level even more, I could lower the design heating load to the point where a central heating system would not be required. By the same token, air conditioning would not be necessary. The cost savings resulting from not having to install a central heating and air conditioning system readily offset the increase in construction costs associated with the extra insulation.

The small amount of energy required for space conditioning and the low cost of electricity made electric baseboard heating a viable alternative to central heating and air conditioning. Advantages of this system include

Energy Efficiency Features

Windows

- Low-e argon gas-filled windows: $U = 0.34$, solar heat gain coefficient (SHGC) = 0.33, visible transmittance (VT) = 0.53.
- Leakage rate (windows and sliders): 0.25 CFM/ft².
- Orientation of windows: south 41%, east 27%, west 32%, north 0%.
- Total window area: 15% of floor area.
- Roof overhang: 28 inches. For the south windows, this reduces solar gain during the summer but permits substantial solar gain in the winter.
- Daylighting and cross ventilation: open floor plan in living/kitchen areas; all rooms have windows (casement), with the exception of a small bathroom.

Exterior Doors

- Metal with polyurethane insulation.
- Front door (two glass half panels): $U = 0.31$, leakage rate = 0.01 CFM/ft².
- Side door to garage, studio/laundry room door to garage, exit door from garage (all solid panels): $U = 0.17$, leakage rate = 0.01 CFM/ft².

Envelope

- Exterior walls: advanced framing (2 x 6 studs 24-inches OC, two stud corners, single top plate, all unnecessary cripples eliminated, single 2 x 12 headers bearing on notched studs, no trimmer studs, no headers in nonsupport walls, drywall clips, etc.). Insulation: R-19 unfaced fiberglass batts in cavities and R-10 extruded polystyrene rigid foam applied over OSB sheathing. Opaque wall composite R- value of 29.
- Raised heel trusses 24 inches OC with 28-inch overhang.

- Ceiling: R-60 blown-in cellulose, correct settled depth verified.
- Air sealing: interior - airtight dry wall approach (ADA); exterior - taped extruded polystyrene sheathing, natural air leakage rate = 0.08 ACH.
- Vapor retarder: Inside wall surface - “vapor retarder” primer paint.
- Drainage plane: Taped extruded polystyrene sheathing.

Foundation/Floor

- Frost-protected shallow foundation monolithic slab.
- Insulation: Extruded polystyrene; R-20 on the outside wall, R-10 extending inward under the footer and floor for 4 feet, R-5 under the remainder of the floor.

Domestic Hot Water

Electric: 40-gallon tank, 0.88 EF rating; centrally located in a bathroom closet; inlet and outlet pipes insulated.

ease of zoning, negligible maintenance costs, space savings, simplicity, reliability, low installation costs, lack of any outside equipment, safety (no combustion products), and energy billing from only one supplier.

Although electric-resistance heating is the most practical and economical choice in this case, it does carry a penalty in terms of the primary energy source. Unfortunately, the co-op consortium does not generate or purchase any green power, nor does the local co-op have a program whereby members can choose to purchase some or all of their electricity from green power sources. However, both of these options are being evaluated and may become available in the future. Also, the house is oriented in such a way that part of the roof area faces due south. This provides a platform for the installation of PV and solar water-heating systems, should these options become viable in the future.

I chose the frost-protected shallow foundation technique for the building because it costs less, saves time and materials, and is less complex than a separate footer, foundation, and slab system. The code requirement for my area is for the edge to be thickened to 18 inches (6 inches above and 12 inches below grade), and for 1 inch of extruded polystyrene foam board insulation to be installed vertically on the outside wall. I upgraded the insulation to a thickness of 4 inches on the outside vertical wall—only the inner 2 inches was installed at the time that the slab was poured—with 2 inches of insulation under the thickened edge and floor extending to a distance of 4 feet inward,

and 1 inch under the remainder of the slab (see Figure 1, and photo, p. 36).

I used the advanced framing technique (minimizing the use of wood

joints) as the covering on wall sheathing because it eliminates thermal bridging, adds R-value, keeps the interior of the wall warmer (and thus acts as an extra safeguard against condensation), and functions as a very effective drainage plane. I increased the thickness to 2 inches for this house. The foam board runs from the bottom of the foundation to the vent chutes in the attic. In addition to added insulation above the top plate area, the rigid foam board also serves as the dam for the blown-in ceiling insulation (see Figure 1). The outside surface of this foam board that is between grade and the bottom plate (the exposed portion) is coated with a fiberglass-reinforced cement coating applied over a self-sticking plastic mesh.

I prefer the airtight drywall approach (ADA) for air sealing to achieve a tight house. I personally did all the air sealing work except that directly related to the gypsum board installation, using a minimally expanding, preblended, single-component polyurethane foam from a 16-lb canister and about 20 tubes of high-quality latex sealant as appropriate.

We designed the garage based on several important criteria. First, the garage has to provide room for storage as well as two cars. Second, the garage is the logical location for the well pump pressure tank. Third, part of the garage space is dedicated to a combination studio/laundry room. Fourth,

finished attached garages are standard for housing in this region.

Therefore, I specified a 24 ft x 36 ft (864 ft²) garage. The 215 ft² studio/laundry room portion of the garage is conditioned space and is located in a back corner. The garage is constructed similarly to the house in terms of framing, insulation, and air sealing with the

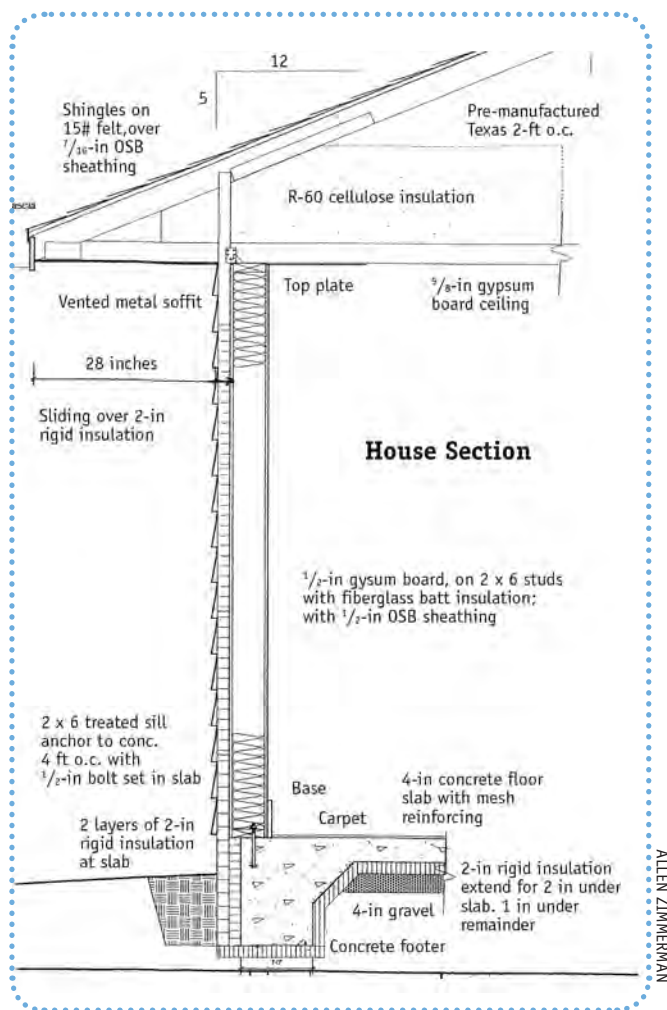


Figure 1. In addition to added insulation above the top plate area, the rigid foam board also serves as the dam for the blown-in ceiling insulation. The bottom plate is cantilevered approximately 1 1/2 inches (the OSB sheathing accounts for the other 1/2 inch) so that the exterior surface of the second layer of 2 inch rigid insulation on the foundation wall aligns with the exterior surface of the 2 inch rigid insulation on the frame wall creating a continuous plane from footer to vent chute.

members not needed for structural support) to frame the exterior of the house (see photo, p. 39). The exterior walls consist of 2 x 6 studs 24-inches on center (OC), with unfaced fiberglass insulation in the cavities. The trusses are stacked above the studs so that a single top plate could be used. I prefer to use extruded polystyrene foam board (taped at the

exception of the amount of insulation. In the laundry area, I installed a tight-fitting floating shuttle cup dryer vent closure instead of the typical leaky flap-per-type dryer vent hood.

Due to the house design and roof framing, the attic access hatch could be located in the garage ceiling. Since the garage is insulated and air sealed, I

a design heating load of 9.7 Btu/ft². The calculated seasonal heating load translates into an annual cost for electrical heating of about \$350. I estimated an additional 10% annual heating load and cost for heating the studio/laundry room and the garage (to keep its air temperature above freezing). Don Jones (at the time owner of DMJ Residential

A 70 CFM fan in each bathroom with a 0.5 sone sound rating, 15W power consumption, and tight-fitting backdraft damper, and a standard range hood in the kitchen provide spot ventilation. Short lengths of metal duct run under the attic insulation and vent the fans to the outside through the soffit. The backdraft damper that came with the range hood was not well made; therefore, a spring-loaded damper was installed in the ductwork instead. The family uses a mechanical crank timer to activate the bathroom fans.

Whole-house ventilation is provided by a system designed to operate intermittently. A 190 CFM fan with a 1.5 sone sound rating, 32W power consumption, and tight-fitting backdraft damper is located in the hallway ceiling. This fan is also vented to the outside through the soffit. (Strong winds did occasionally defeat the backdraft damper in this fan, so I retrofitted a spring-loaded damper in its ductwork.) The fan is wired in parallel to a dehumidistat and a simple 24-hour timer. Whenever the humidity in the house gets too high, the dehumidistat automatically operates the exhaust fan. The timer can also be set to operate the fan during selected periods of the day.

Fresh air is brought into the house through a 6-inch metal duct running from a soffit inlet to ceiling grilles located in the bedroom closets and wall grilles located in the family and living rooms. The air inlet duct runs under the attic insulation and contains a cape-type backdraft damper that opens when the fan is operating. The closet and bedroom doors are undercut to allow the incoming air to move throughout the house.

Plumbing Considerations

Some of the design decisions for locating the bathrooms, kitchen, and laundry room involved minimizing lengths of plumbing runs. In addition, I located the electric water heater in a closet in one of the bathrooms to conserve water and energy (hot water is quickly available in the bathrooms). I also insulated



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This wall section illustrates the use of advanced framing techniques. Notice the single header on the window openings that made room for insulation installation, while saving wood.

designed and installed a sealed and insulated hatch (see “Attic Accesses: High Priority or No Big Deal?” *HE* Mar/Apr '05, p. 16).

Heating System

I developed a simplified, holistic approach to introducing, explaining, and estimating the design heating load, seasonal heating load, and heating performance of houses for use in my courses (“A Primer on Estimating Heat Loads and Performance”). Based on calculations using this method, the design heating load for the house (minus the studio/laundry room) is approximately 19,000 Btu/hr, or 5,500 watts, and the seasonal heating load is approximately 20 million Btu or 6,000 kWh. This yields a house heating index of 1.7 Btu/ft²/HDD (heating degree-day) and

Building Analysis, Columbus, Ohio and now deceased) used REM/Rate software to evaluate the predicted house performance and obtained values similar to those I derived.

A total of 32 feet of baseboard heater capacity (nine heaters varying from 2 to 5 feet each with a dedicated wall thermostat) was installed in the house. A 4-ft baseboard heater was installed in the studio/laundry room and a 6-ft heater in the garage. Because of the well-insulated and tight construction, I was able to locate the heaters on interior walls to save on installation costs and reduce openings in the drywall in exterior walls.

Ventilation System

I designed a simple controlled-exhaust ventilation system for the house.

the pipe adjacent to the water heater. Since the heater is located in the conditioned space of the house, any energy lost during the heating season simply replaces electric energy that would have been used directly for heating.

I prefer cross-linked polyethylene (PEX) tubing for plumbing. Only drain, waste, and vent (DWV) piping is located under the floor slab. All supply piping runs above the ceiling or in the walls. The pipes are embedded in the wall and ceiling insulation toward the interior side, which means that a considerable amount of insulation and R-value separates the pipes from the outside. Standard commercially available water-conserving appliances and fixtures are used in the house.

Lighting, Appliances, and Miscellaneous Plug Loads

The main lighting fixtures in the kitchen and bathrooms use energy-efficient fluorescent lamps. CFLs serve as replacements for incandescent lightbulbs in floor and table lamps that are operated frequently and for long periods of time. We purchased all the new appliances from a local dealer with whom we have a longstanding relationship. At the time (2000) this business did not inventory any Energy Star appliances appropriate to our needs. I expect that the situation has now changed. As *Home Energy* readers know, appliance energy use depends on both the appliance design and the management of its use. Since Susan and I know how to use appliances efficiently and put this knowledge into practice, I was comfortable with the decision to buy appliances that are efficient, but not up to Energy Star standards.

Miscellaneous plug loads, such as entertainment centers, DVD players, and TIVO, can use up to 15% of the average home's electricity. As homes become more efficient at heating, cooling, and making hot water, these loads become much more significant (see "Roadblocks to Zero-Energy Homes," p. 24). Our only appliances that use standby power are a voice mail recorder and a

small LCD TV/DVD combination unit; both have an Energy Star rating.

Building Performance

Don Jones performed a blower door test on the house with the fresh-air intake system open and measured a CFM₅₀ rating of 667. Based on this and on the building parameters, Jones estimated a natural air leakage rate of only 0.13 ACH. The natural air leakage rate of the building shell (with the air intake system closed) was estimated to be only 0.08 based on extrapolation.

windows. Therefore, we maintain the RH at 40% during the coldest periods of the winter and 45%–55% during the rest of the heating season. The limited amount of condensation does not cause any serious problems and soon evaporates during the day.

The house is located in a region in which the typical number of heating degree-days is 6,160 and the winter design temperature is 1°F. Based on the HDD value, two of the winters have been colder and four have been warmer than normal. The lowest temperature recorded during the six-year period has

Table 1. Annual Heating Load and Cost Information

Heating Season	00-01	01-02	02-03	03-04	04-05	05-06
Heating degree days (HDD)	6,424	5,503	6,528	6,118	6,128	5,662
HDD Deviation (6,160 Base)	+4.3%	-10.7%	+6.0%	-0.7%	-0.5%	-8.1%
Heating Load (kWh)	6,600	4,500	7,000	6,200	5,700	5,300
Electric Rate (\$/kWh)	\$0.053	\$0.057	\$0.057	\$0.06065	\$0.06065	\$0.06065
Total Heating Cost	\$350	\$255	\$395	\$375	\$345	\$320

We have lived in the house for six years (move-in date was mid-October 2000), and it has met all our expectations regarding design criteria. We have found that the house is comfortable at a steady-state indoor temperature of 65°F. Thermostats are set slightly higher in some rooms when those rooms are occupied. Thermostat settings are kept lower or turned off in rooms that are not typically occupied.

I found that the combination of natural air leakage via the thermal shell and the air intake system, proper operation of the bathroom fans, and controlling the central exhaust fan via the dehumidistat provides adequate ventilation without the need to activate the central exhaust fan using the 24-hour timer. The relative humidity (RH) in the house during the winter is maintained within the comfort range of 40%–55% and is adjusted based on outside temperature to control window condensation. The spacer strip between the double panes of glass is a thermal weak point, and condensation will first occur at this strip on the bottom of the

been minus 5°F. The temperature in the garage has never dropped below 45°F during any of the six winters; therefore, no supplemental heat has been required.

Based on the amount of electrical energy metered during the autumn and spring months when little or no supplemental heating is required, I estimated that the typical kWh of electrical energy metered monthly for nonheating electrical use during the winter months is about 600 kWh (0.28 kWh/ft²) per month for my home. This is very low compared to representative values for shoulder months derived from data in a 2001 Partnership for Advanced Technology in Housing (PATH) report for new all-electric homes in central and northern Ohio (0.5 kWh/ft² per month and 0.6 kWh/ft² per month, respectively). I subtracted this value from the actual kWh metered for each of the months during the heating season to determine the kWh required for supplemental heat of the 2,165 ft² of conditioned space.

HDD, heating kWh, and heating cost information for six heating seasons are

shown in Table 1. Since the house is not air conditioned, the heating system does not include a fan to move air, and the baseboard heaters require no maintenance, the values listed represent system totals for space conditioning the house. The average total annual energy use and costs for space conditioning are 5,890 kWh (2.7 kWh/ft²) and \$340 respectively.

The six summers that the house has been occupied have provided a substantial test of the performance of the building in the absence of air conditioning. The house design features have kept the indoor environment very comfortable during typical summer conditions. Even under conditions of very high temperature and RH (which occur a few times each year), the combination of a well-insulated and tight house and the daily management of opening windows, adjusting window shades, and using portable fans have kept us comfortable indoors. In the future, when the deciduous shade trees that we planted on the south and west sides of the house have grown larger, the house will perform even better.

Jones also performed an energy analysis of the house using REM/Rate software and came up with a rating of 82.9. A rating of 86 is required for Energy Star designation. The rating system carries a heavy penalty for the electric-resistance heat, yet no credit is awarded for the fact that the design of the house eliminates the need for air conditioning and the energy it would consume. As a point of information, electricity use in northern Ohio actually peaks in the summer instead of the winter, due to heavy air conditioning loads.

Jones calculated that even a low-efficiency (80%) AFUE (annual fuel utilization efficiency) propane furnace would

increase the rating to 88.9. However, the system operating (variable) costs to heat with propane would by far exceed the costs for resistance heating. In addition, the initial costs of a propane system would be much higher. Jones also calculated that a 7 HSPF (heating seasonal performance factor) air source heat pump would increase the rating to 88.9. This option would lower the energy costs for heating, but only by about \$150 annually. Given the much higher initial and maintenance costs associated with such a system, the payback for this option would be unrealistically long.



I prefer cross-linked polyethylene (PEX) tubing for plumbing. Only drain, waste, and vent (DWV) piping is located under the floor slab. All supply piping runs above the ceiling or in the walls.

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heat pumps in central and northern Ohio are 9.3 kWh/ft² per year and 11.7 kWh/ft² per year, respectively. Even for homes with gas furnaces in central Ohio (many or all of which would also have gas water heaters), the average total annual electrical energy consumption is 4.4 kWh/ft² per year.

Home, Sweet Home

Building a comfortable and energy-efficient house does not require far-out designs, exotic materials, or complex assembly techniques. However, it does require a good understanding of building science. Design and construction principles and practices should be based on building science and engineering fundamentals, and should take the building-as-a-system approach. The well-insulated, tight, and correctly ventilated house that results will be comfortable, safe, healthy, durable, quiet, practical, affordable, and energy efficient.



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In short, I find the Energy Star criteria, as applied to this very energy-efficient house and the available energy sources, disappointing in two major respects. First, the ratings are biased toward heating systems that are clearly not the best practical and economic choices. And second, the ratings do not give any credit to the effect of no air conditioning on the total energy consumption.

The average total annual energy consumption for the five full years that we have lived in our all-electric home (baseboard resistant heaters, no AC) is 11,870 kWh (5.5 kWh/ft²) per year. In comparison, representative values based on data in the PATH report for the average total annual electrical energy consumption for all-electric homes with

For more information:

To learn more about the author's home, or for a copy of "A Primer on Estimating Heat Loads and Performance," contact him at

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